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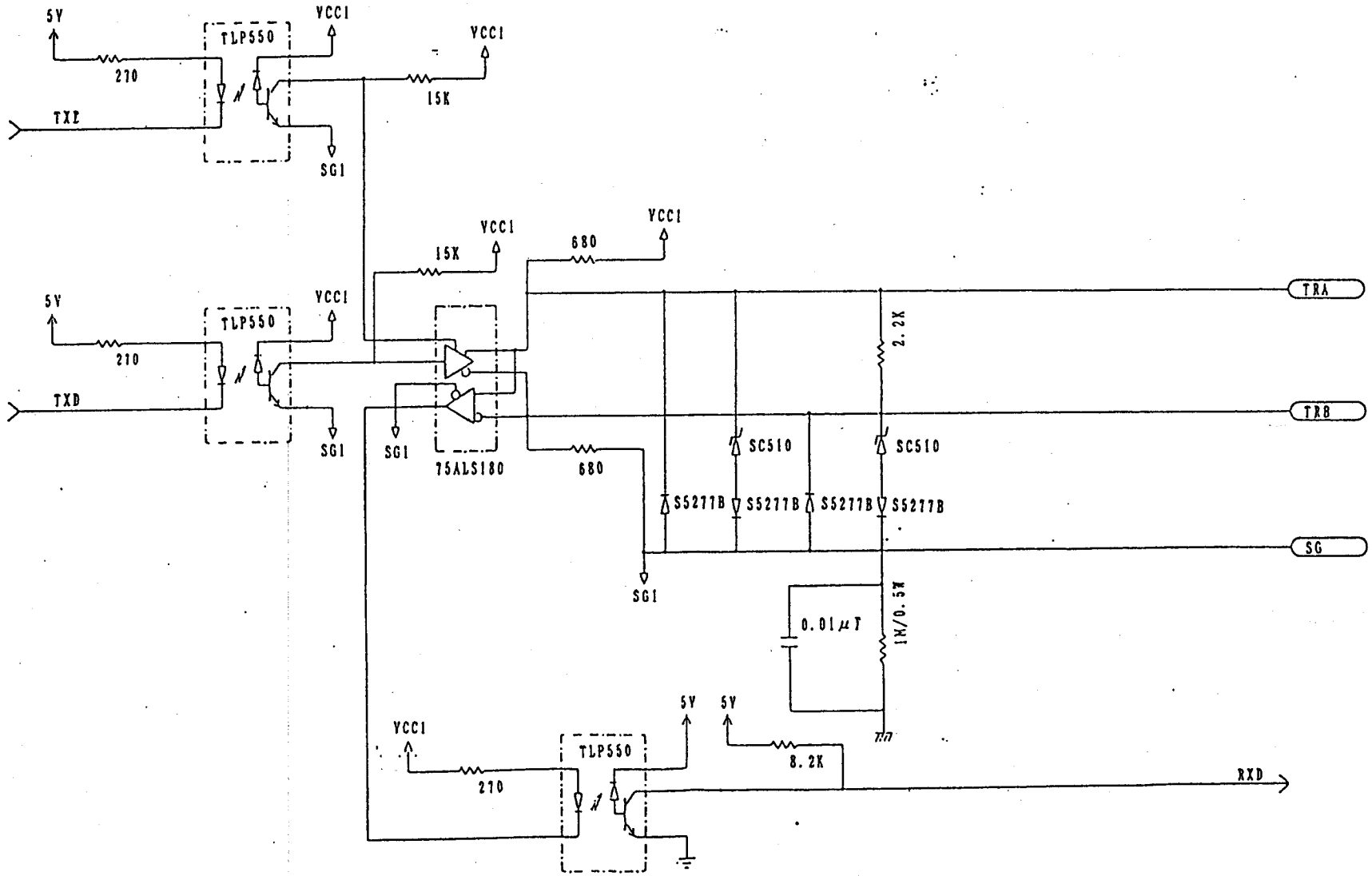
# 1.Input/Output Circuit for Interface Signal

## 1-1.Interface Signal

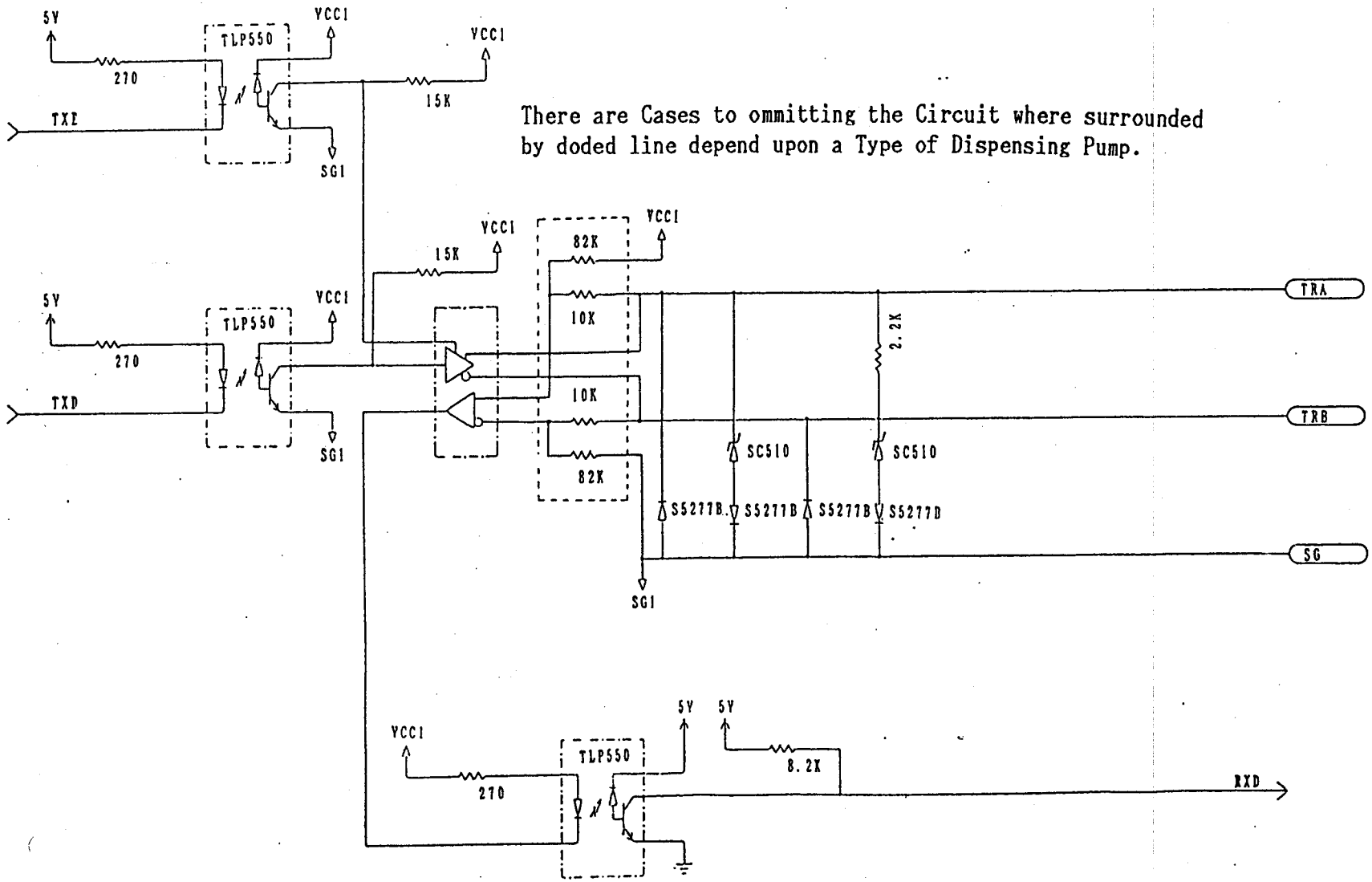
| CIRCUIT SIGNAL FOR INTERFACE | NAME OF CIRCUIT | SIGNAL DIRECTION |
|---|---|---|
| FG | FRAME GROUND | NO PARTICULAR DIRECTION |
| TRA | TRANSMITTING/RECEIVING DATA ( A ) | BOTH DIRECTIONS |
| TRB | TRANSMITTING/RECEIVING DATA ( B ) | BOTH DIRECTIONS |
| SG | SIGNAL GROUND | NO PARTICULAR DIRECTION |

## 1-2.Console Side Interface Circuit



## 1-3.Dispensing Pump Side Circuit

There are Cases to ommitting the Circuit where surrounded by doded line depend upon a Type of Dispensing Pump.



## 2. Transmission Procedure

### 2-1. Basic Specifications

(1) Interface Standard; Confirmed to RS-485 of EIA standard.
(2) Synchronous Method; Asynchronous.
(3) Transmission Speed; 19,200 BPS
(4) Transmission Method; Half Duplex
(5) Transmission Procedure; Control Procedure. (polling/selecting method)
(6) Parity Check; Horizontal(even), Vertical(even) Parity Check.
(7) Correction; Automatic. (ACK-NAK Method) JIS-C-6220, 8unit Symbol
(8) Code; ASCII Code,
(9) Composition of One Character; 11 Bit.

| | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| Start bit | b0 | b1 | b2 | b3 | b4 | b5 | b6 | b7 | Parity bit | Stop bit |

## 2-2.Transmission Control Symboles and Sequence

| Control Symbol and Sequence | Functon | |
|---|---|---|
| Prefix.ENQ. ENQ Sequence) | Polling of the Transmission for dependency. (Polling or Selecting Sequence) | |
| STX | Indication of the Start Text/Block | 02 |
| ETX | Indication of the End Text. | 03 |
| ACK 0<br>ACK 1 | Indication of the Positive Response for the Text and Selecting Sequence.<br>(Alternately response ACK.0/ACK.1) | 10 30<br>10 31 |
| NAK | Indication of the Negative Response for the Text/Block | 15 |
| ENQ | Indication of the Demand response for the Text/Block | 05 |
| EOT | Indication of abandonment of the right of the Transmission. | 04 |

## 2-3.ENQ Sequence

| | | | |
|---|---|---|---|
| E<br>O<br>T | S<br><br>A | U<br><br>A | E<br>N<br>Q |

UNIT ADDRESS

41H : Selecting
51H : POLLING

Station Address

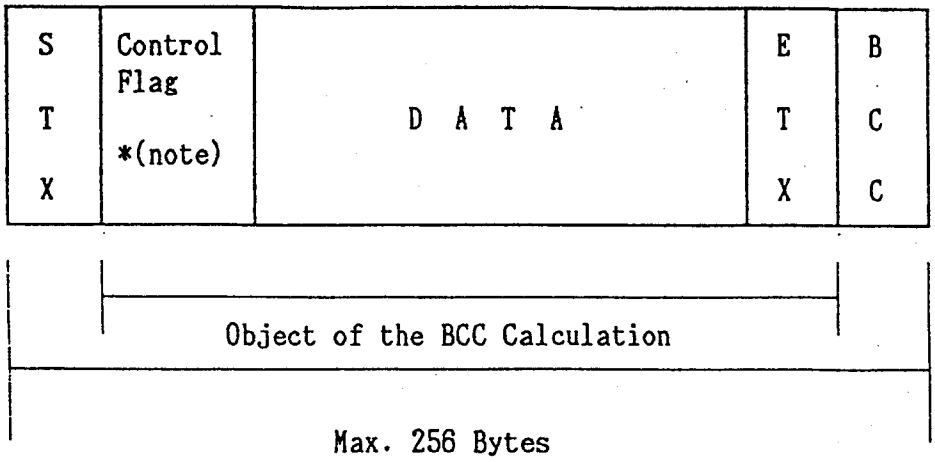
Device No.(40H~7FH)which is assigned to the each Dispensing Pump.

40→41

30×6

7F

## 2-4.Text Format



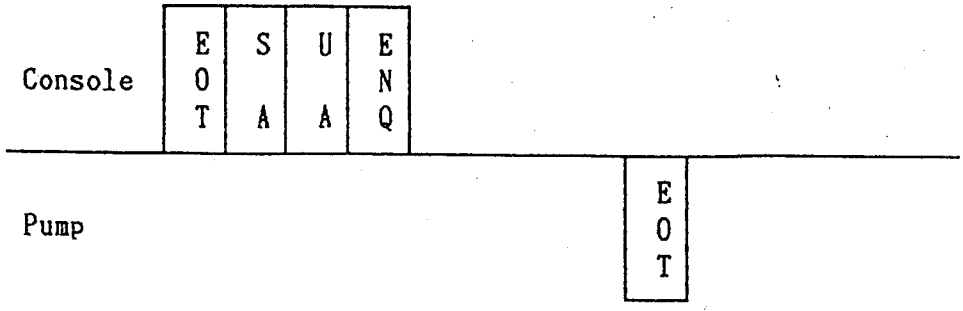

*(Note) : Composition of the Control Flag

| | |
|---|---|
| S<br>A | U<br>A |

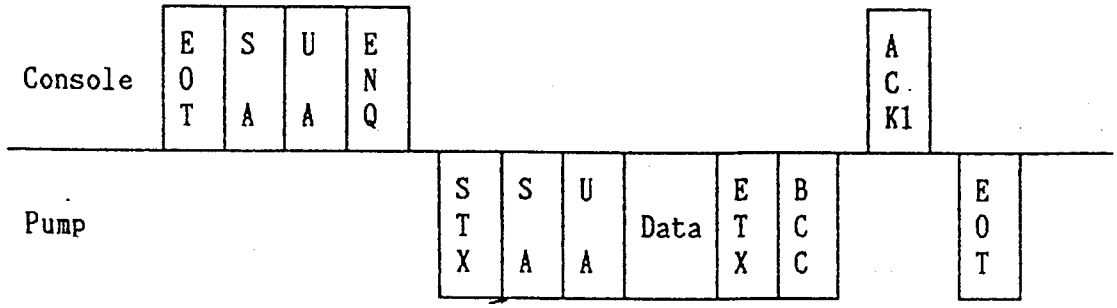
The Contents are the same as the ENQ Sequence.

## 2-5.Example of the Transmission Sequence

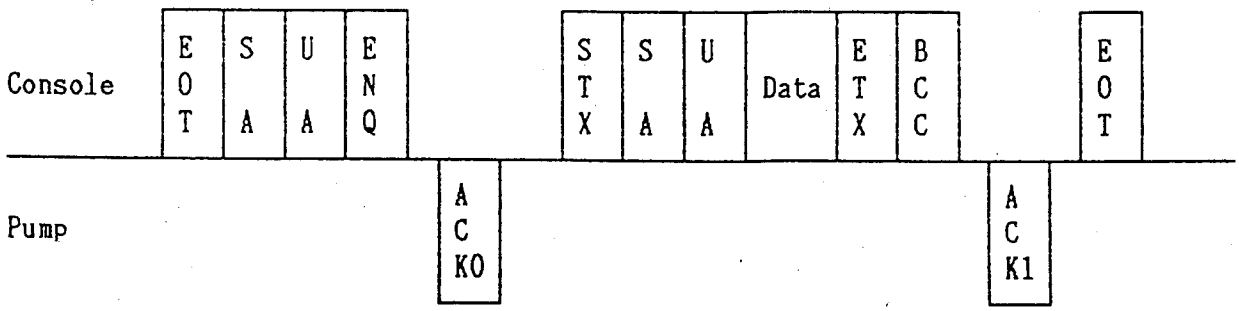
(1).Polling.(In case of no Transmission data is exist)



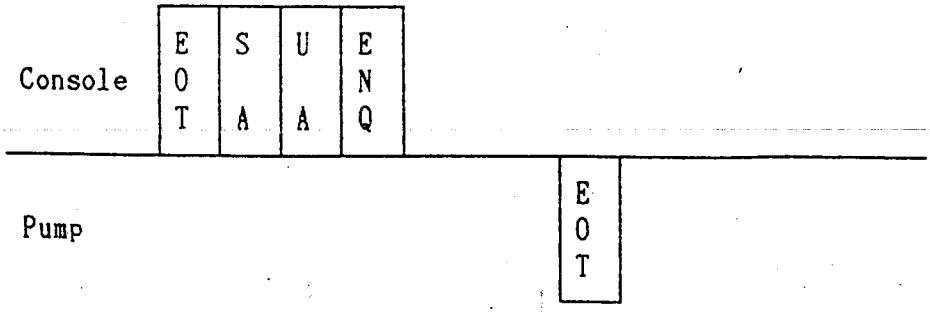

(2).Polling.(In case of the Transmission data is exist)




(3).Selecting.




(4).Selecting.(In case of no data can receive)



## 3.Time Provisions

### 3-1.Interval of Transmission Character in One Block.

(1)From Console to Dispensing Pump ; within 1,7ms.
(2)From Dispensing Pump to Console ; within 1,7ms.

The Time described above are shown Transmittig 32 bit at the baud rate of 19,200 BPS.

When the Time exceeded more than this,it will be handled as Transmission End.

### 3-2.Change over Time for Transmission/Reception at Console Side.

(1)From Transmission to Reception. ; The Time between right After trasmitting Stop bit of Final Character, and ditecting Bigining Characters Start bit of Reception Data.

1.7ms.

(2)From Reception to Transmission ; The Time between right after detecting Stop bit of Final Character and transmit the Start bit of biginig Character.

①Before settle Data-link,
Over 5,0ms and shorter than 500ms.

②After settled Data-link,
Over 5,0ms and shorter than 500ms.

### 3-3.Wait Time for the Response at Console side.

Longer than 500ms.

## 4.Transmission matrix

### 4-1.Polling at Console Side.

| | ① Wait for response after Transmit the POL.ENQ | ② Wait for the Text after transmit the response |
|---|---|---|
| TEXT | BCC:OK- Send ACK1 and move to ②<br>BCC:NG- Send NAK and move to ② | BCC:OK- Send ACK1 and move to ②<br>BCC:NG- Send NAK, if the C1 is within five times,and move to ②<br>In case Retry over,move to Idle ① |
| EOT | Move to Idle ① | Move to Idle ① |
| TIME OUT | Move to Idle ① | Move to Idle ① |

4-2.Selecting at Console side

| | ① Wait for the response after Transmit the SEL.ENQ | ② Wait for the Response after transmit the TEXT | ③ Wait for the Response after transmit the Demanded ENQ |
|---|---|---|---|
| ACK0 | Transmit the TEXT and move to ② | Move to Idle ① | Retransmit the TEXT |
| ACK1 | | Move to Idle ① | Move to Idle ① |
| NAK | | Within five times of CI,Retransmit the TEXT and move to ②<br>Retry over:<br>Move to Idle ① | Within five times of CI,Retransmit the TEXT and move to ②<br>Retry over:<br>Move to Idle ① |
| EOT | Move to Idle ① | | |
| TIME OUT | Move to Idle ① | Within five times of CI,etransmit the ENQ and move to ③<br>Retry over:<br>Move to Idle ① | Within five times of CI,etransmit the ENQ and move to ③<br>Retry over:<br>Move to Idle ① |

4-3. Polling at Pump side.

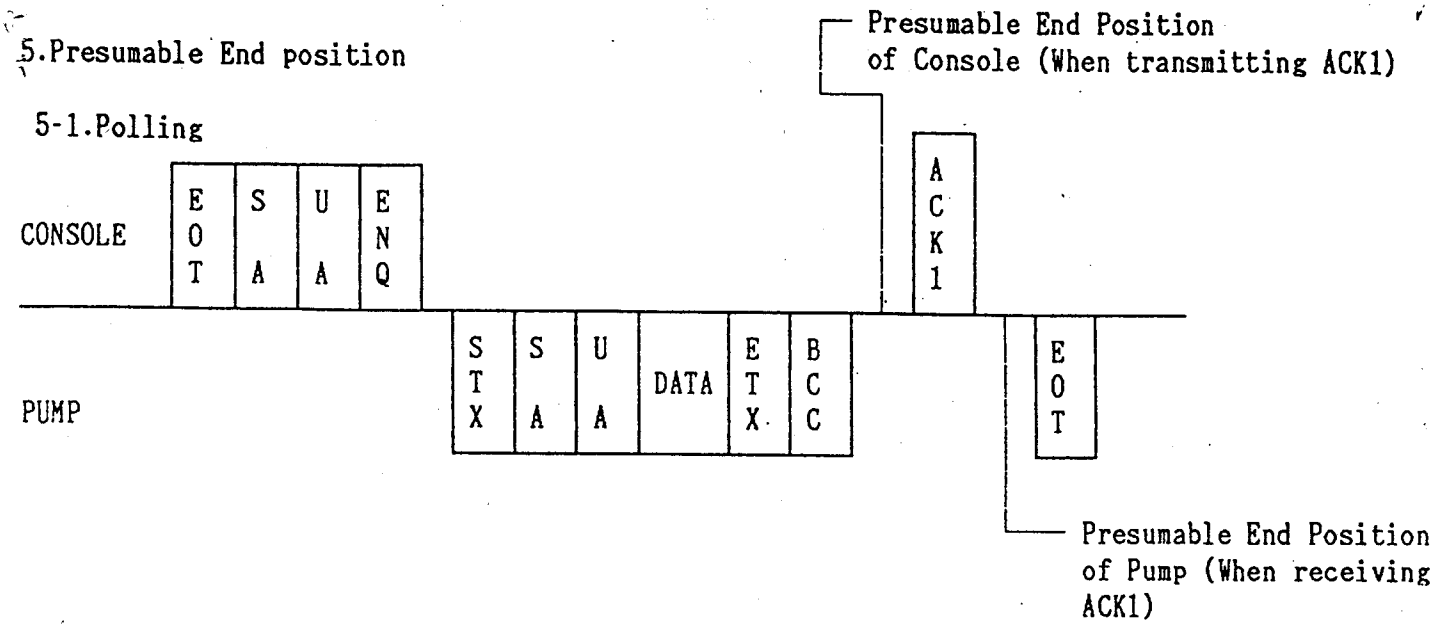
| | ① Wait for POL.ENQ/SEL.ENQ | ② Wait for the response after transmit the TEXT |
|---|---|---|
| POL. ENQ | SA.UA:OK- Transmit the TEXT and move to ② or transmit the EOT and move to ①<br><br>SA.UA:NG- Move to ① | SA.UA:OK - Retransmit the TEXT and move to ②<br><br>SA.UA:NG - Move to ① |
| ACK1 | | Transmit the EOT and move to ① |
| NAK | | Within five times of CI,retransmit the TEXT and mone to ②<br>Retry over: Transmit the EOT and move to ① |

4-4. Selecting at pump side.

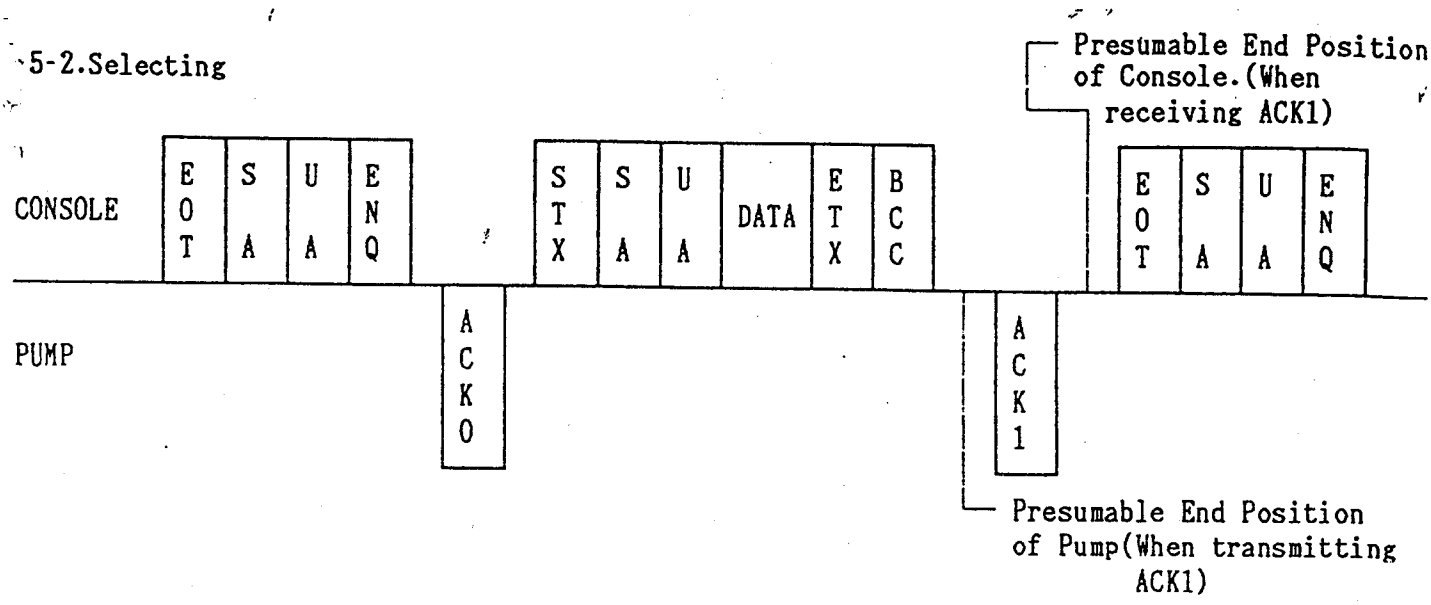
| | ①Wait for the POL.ENQ/SEL.ENQ | ②Wait for the TEXT | ③Wait for the response after transmit the ACK1 |
|---|---|---|---|
| SEL. ENQ | SA.UA :OK- Transmit the ACK0 and move to ② or transmit the EOT and move to ①<br>SA.UA:NG-Move to ① | SA.UA : OK-Transmit the ACK0 and move to ②<br><br>SA.UA : NG-Move to ① | SA.UA : Transmit the ACK0 and move to ② or transmit the EOT and move to ①<br><br>SA.UA : NG- Move to ① |
| TEXT | | BCC : OK -Transmit the ACK1 and move to ③<br><br>BCC : NG -Transmit the NAK and move to ② | |
| ENQ | | Retransmit the just previous response and move to ② | Retransmit the just previous response and move to ③ |

## 5.Presumable End position

### 5-1.Polling



5-2.Selecting




(NOTE)
At the POLLING/SELECTING,in case,the Pump/Console did not receive ACK1 for the TEXT, the Pump/Console shall re-transmit the TEXT at the Next Sequence.

Care must be taken about the possibility to have duplicate TEXT for both polling and selecting when communication error with the above mentioned communication sequence is occured.

Especially, for Console,it must be provided with some measure which wil not take a fuelling data twice when its receives duplicate fueling status.

## 6.TEXT FORMAT

### 6-1.Transmission TEXT from Console to Pump.

1). Fueling authorization (1).
2). Fueling authorization (2).
3). Cancelation of fueling authrization.
4). Pump lock.
5). Release of pump lock
6). Request of Dispensing pump status.
7). Request of totalized value of Dispensing Pump.
8). Sub unit price indication data

6-1-1. Fueling authorization (1)

| TEXT discrimination code (2) | Terms of authorization (1) | Volume / Amount (1) | Volume or Amount ( 4+2 ) (6) | U.price flag (1) | U.price (4) |
|---|---|---|---|---|---|

'10'

0 : Normal fueling (see NOTE 2)
1 : Fueling with volume limitation
2 : Preset fueling (Preset value can be changed at Dispensing Pump)
3 : Preset fueling (preset value cannot be changed at Dispensing Pump)

1 : Volume
2 : Amount

0: Prohibition of fueling
1: Fueling authorization (Unit prcice invalid)
2: Fueling authorization (Cash unit price) (NOTE 4)
3: Fueling authorization (Credit unit price) (NOTE 4)

9: Fueling authorization (Unit price selected with dispensing pump) (NOTE 5)

(NOTE 1) Location of decimal point for Volume,Unit price and Amount are the same as Display.
(NOTE 2) Normal fueling,
Authrize fueling from console without designate Volume and Amount.
In this case, limited volume will be applied from Dispensing Pump.
(NOTE 3) This TEXT will not be applied for MPP type Dispensing Pump.
(NOTE 4) Handled as a cash unit price (unit price flag = 2) and as a credit unit price (unit price flag = 3).
(NOTE 5) When the unit price (unit price flag = 9) selected with a dispensing pump is designated, the unit price becomes invalid and is not used.

6-1-2. Fueling authorization (2)

| TEXT discrimination code (2) | Terms of authorization (1) | Volume / Amount (1) | Volume or Amount ( 4 + 2 ) (6) | Product 1 Flag (1) | Product 1 U.Price (4) | Product 2 Flag (1) | Product 2 U.Price (4) | | Product 6 Flag (1) | Product 6 U.Price (4) |
|---|---|---|---|---|---|---|---|---|---|---|

'11'

1 : Volume
2 : Amount

0 : Normal fueling (NOTE 2)
1 : Fueling with volume limitation
2 : Preset fueling (Preset value may be controlled from Pump)
3 : Preset fueling (Preset value may not be controlled from Pump)

0: Prohibition of fueling
1: Fueling authorization (Unit prcice invalid)
2: Fueling authorization (Cash unit price) (NOTE 4)
3: Fueling authorization (Credit unit price) (NOTE 4)
9: Fueling authorization (Unit price selected with dispensing pump) (NOTE 5)

(NOTE 2) Normal fueling,
Authrize fueling from console without designate Volume and Amount.
In this case, limited volume will be applied from Dispensing Pump.
(NOTE 3) This TEXT will not be applied for MPP type Dispensing Pump.
(NOTE 4) Handled as a cash unit price (unit price flag = 2) and as a credit unit price (unit price flag = 3).
(NOTE 5) When the unit price (unit price flag = 9) selected with a dispensing pump is designated, the unit price becomes invalid and is not used.

- Subject : The TEXT of fueling authrization

(1) It is possible to send them several times while the nozzle is stored in the holster In this case, the last TEXT will be treated as valid one.

2) It is possible to send them out before establishment of fueling,even if the nozzle has been taken out.

~~3)~~ The fueling authorization must be sent after confirmation of the nozzle out by the dispensing pump status in case of the system which is automatically authorize the fueling as soon as nozzle out. for Multi wire dispenser

4) It is treated as the cancelation of the fueling, if the nozzle is restored without any fueling under the condition of authorized fueling. Cancelation command is added

~~5)~~ When the unit price flag is the unit price (unit price flag = 9) to be selected with the dispensing pump, fueling becomes possible only after selection of the unit price rank with the dispensing pump. The unit price sent (sub-unit price indication data text) be forehand to the dispensing pump is used.
Note however that the cash unit price becomes valid when the unit price selector switch is not provided.

~~6)~~ When the sub-unit price indication is provided, the sub-unit price indication of the products not authorized for fueling is invalid indication [-----] for both cash and credit.

6-1-3. Cancelation of fueling authorization

| TEXT Discrimination Code |
|---|
| (2) |

'12'

(NOTE 1) It is not treated as error, if it is no authorized condition.

(NOTE 2) It is treated as pump lock, if it is authorized condition with the nozzle has been taken off.( even it is under the fueling )

6-1-4.Pumplock

| TEXT discrimination Code |
|---|
| (2) |

'13'

- This TEXT is always valid.

.. Release of Pump lock

| |
|---|
| TEXT discrimination Code (2) |

'14'

- This TEXT is always valid.
  However,it will be ignored unless fueling authorized condition.

6-1-6. Request of dispensing pump status

| |
|---|
| TEXT discrimination code (2) |

'15'

- This TEXT is always valid.
  However, it must be controlled not to sent the request again until console receives the status from dispensing the pump after transmission of the status request.

6-1-7.Request of totalization value of dispensing pump

| |
|---|
| TEXT discrimination code (2) |

'20'

- This TEXT is always valid.

6-1-8. Sub-unit price indication data

| Text discrimination code (2) | Unit rank code | Product 1 | | Product 2 | | | Product 6 | |
|---|---|---|---|---|---|---|---|---|
| | | Flag (1) | U. price (4) | Flag (1) | U. price (4) | | Flag (1) | U. price (4) |
| '21' | | | | | | | | |

0: Indication blank
1: Invalid indication [-----]
2: Unit price valid

2: Cash
3: Credit

● This text is always valid.

● Sub-unit indication data text

(1) The sub-unit price indication is renewed by the text sent even during fueling.

(2) The last text sent becomes valid and indicated while the nozzle is put in the holder.

(3) When the nozzle is taken out, the invalid indication is made for indication other than the product sub-unit price indication.

(4) Only the designated product and sub-unit price of unit rank are indicated during fueling.

6-2. Transmission TEXT from dispensing pump to console.

(1) Transmission of the dispensing pump conditions.
(2) Dispensing pump status.
(3) Error message for received TEXT.
(4) Totalization value of dispensing pump.
(5) Switch status

## 6-2-1. Transmission of the dispensing pump condition

| TEXT discrimination code (2) | Condition (1) | Contents (1) |
|---|---|---|
| '60' | | |

All the time '0'

1 : Power ON ( Dispensing pump is controllable )
2 : Dispensing Pump is controllable.
3 : Dispensing pump is uncontrollable. ( NOTE )

( NOTE ) When Dispensing pump is uncontrollable, make it not to send any TEXT to dispensing pump. However,it is allowed to send the request of totalization value and the reqest of dispensing pump status.
It will be back the error message (timing error) for the TEXT.

## 6-2-2. Dispensing pump status

0:nozzle is restored
1:Nozzle (1)
2:Nozzle (2)
3:Nozzle (3)

| TEXT discrimination code (2) | Condition (1) | Fueling volume (4+2) | U.Price flag (1) | U.Price (4) | Amount (6) | Nozzle number (1) | Products (1) | |
|---|---|---|---|---|---|---|---|---|
| "61" | | | | | | | | |

1 : Unit price invalid(U.price and Amount will be '0' )

2 : Unit price valid.

Type of indication (1)

'2'

0 : nozzle is restored/no function
1 : H (High-Octane,Unleaded)
2 : R (Regular,Unleaded)
3 : D (Diesel)
4 : K (Kerosene)
5 : X (Leaded high Octane)
6 : z (Lesded regular)

0 : Nozzle is kept in the holster.
1 : Nozzle is taken out
3 : During fueling
4 : Fueling completion (Nozzle is restored)

( NOTE 1 ) location of decimal point for Volume,U.price and Amount are the same as indication

( NOTE 2 ) This TEXT will be transmitted following conditions,

- In case of changing the Item of 『CONDITION』.
- When received 『REQUEST OF DISPENSING PUMP STATUS』.

6-2-3. Error message of receive TEXT

| TEXT discrimination code (2) | Discrimination code for received data (2) | Error code (1) |
|---|---|---|
| '62' | | 0 : Error of set up data<br>1 : Timing error |

6-2-4. Totalization value of dispensing pump

| TEXT discrimination code (2) | Nozzle 1 (1) | Nozzle 2 (1) | Nozzle 3 (1) | Spare:20 (3) | Nozzle (1) totalization value volume (8+2) | Amount (10) | Nozzle (2) totalization value Volume | Amount |
|---|---|---|---|---|---|---|---|---|
| '65' | | | | | | | | |

| Nozzle (3) totalization value Volume | Amount |
|---|---|
| | |

(Variable)

20H : no Nozzle
0 : Unknown Product
1 : Product 1 (H)
2 : Product 2 (R)
3 : Product 3 (D)
4 : Product 4 (K)
5 : Product 5 (X)
6 : Product 6 (Z)

9 : Error of the totalization value
(Totalization value will be transmitted as 0)

( NOTE 1 ) Location of decimal point for volume and Amount are the same as Indication part.
( NOTE 2 ) Due to variable length, this TEXT will be varied by number of Nozzle.

● In case Number of Nozzle = 1

| Text discrimination code (2) | Product for Nozzle 1 (1) | Spare : 20H (5) | Nozzle (1) totalization value Volume \| Amount |
|---|---|---|---|

● In case Number of Nozzle = 2

| Text discrimination code (2) | Product for Nozzle 1 (1) | Product for Nozzle 2 (1) | Spare : 20H (4) | Nozzle (1) totalization value Volume \| Amount | Nozzle (2) totalization value Volume \| Amount |
|---|---|---|---|---|---|

● In case Number of Nozzle = 3

| Text discrimination code (2) | Product for Nozzle 1 (1) | Product for Nozzle 2 (1) | Product for Nozzle 3 (1) | Spare : 20H (3) | Nozzle (1) totalization value Volume \| Amount | Nozzle (2) totalization value Volume \| Amount | Nozzle (3) totalization value Volume \| Amount |
|---|---|---|---|---|---|---|---|

## -2-5 Switch status

| Text discrimination code (2) | Switch code (1) | Product code (1) | Designation code (1) |
|---|---|---|---|
| '67' | | | |

Designation code (1):
- 1: Rank designation cancel
- 2: Cash designation
- 3: Credit designation

Product code (1):
- 0: No designation (Note 1)
- 1. Product 1
- 2. Product 2
- 3. Product 3
- 4. Product 4
- 5. Product 5
- 6. Product 6

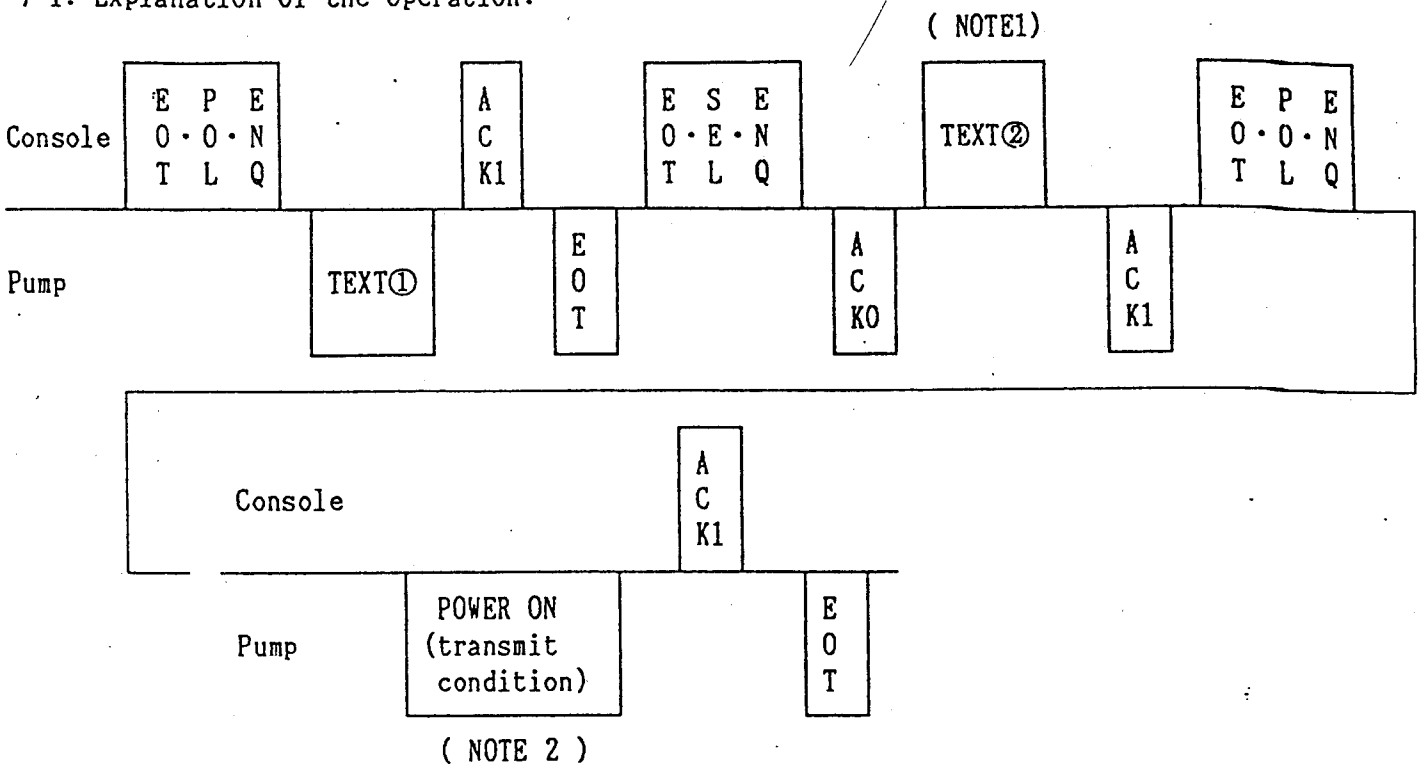
Switch code (1):
- 1: Unit price selector switch
- 2: Spare
- . Spare
- . Spare
- . Spare
- 9: Spare

(Note 1) When a unit price selector switch exists for each rank, designation of the rank while the nozzle is in the holder does not settle the product, resulting in "No designation".

## 7. Operation procedure when Power is ON.

Dispensing pump will transmit the "POWER ON" TEXT after conducted the sequence below,during power is ON, then will start it's operation under the control by the Console.

### 7-1. Explanation of the operation.




( NOTE 1 ) Console will operated for the contents of the TEXT (1) from dispensing pump, then transmit the results TEXT.

( NOTE 2 ) Dispensing pump will transmit the "POWER ON" TEXT,If operated results are tally both dispensing pump and console.
If does not tally the operated results,the dispensing pump will retransit the TEXT(1).
In case, unsucceeded the operated results within three(3) times, the console could not be able to control the dispensing pump.

## 7-2.Explanation of the Transmission TEXT.

Dispensing pump → Console

| TEXT ID CODE | DATA (1) |
|---|---|
| (2) | (4) |

'00'

Operate area of ~~CRC~~ XOR

Console → Dispensing pump

| TEXT ID CODE | DATA (2) |
|---|---|
| (2) | (2) |

'00'

(1).Operate the CRC
Created multiple formula

$$x^{16} + x^{12} + x^{5} + 1$$

(2).Addition for the upper bytes and lower bytes of the CRC.

(3).Convert to the unpack data.

(4).Transmit the operated results to the dispensing pump.

Example

2081H

A1H

4131H